

TranscriptShift: study-disjoint and site-disjoint asthma-expression transfer audits

Research report, September 25, 2026. This document is a candidate-specific expansion of the short PAPER.md, not a chapter counted again in the aggregate item-27 manuscript. Its first experiment is a negative test of transfer between two GEO cohorts in one parent publication. Its second is a separate negative within-study site transfer using another GEO series and microarray platform. Neither makes a clinical diagnostic claim.

Abstract

A model that classifies cases in a sample-random partition may rely on study conditions rather than disease signal. We used exact original GEO series matrices for GSE141623 (BAMSE, 256 blood-expression profiles) and GSE141631 (INMA, 201 profiles), two separate cohorts under the same parent GSE141661. Phenotypes were parsed from per-sample asthma: 0/1 characteristics. A panel of the first 1,000 valid probe rows was selected without phenotype inspection. Fixed logistic regression ($C=0.1$) with train-only median imputation and standardization was trained on one full study and evaluated on the other. AUC was 0.457 BAMSE to INMA and 0.521 in reverse, below the build-time exploratory >0.65 two-direction gate. An independent third GEO series, GSE110551 (156 adults across Wroclaw and Zurich), uses a different GPL570 probe space. It therefore cannot validate the first fitted model. A separate, identically specified model trained on one site scored AUC 0.613 on the other in one direction and 0.545 in reverse. No tuned neural model was compared because the fixed linear baseline did not pass. The useful result is the fail-closed split and provenance audit, not a validated asthma classifier.

1. Research question and practical use

The question is whether a simple expression-based classifier holds when an entire study or site is withheld, not whether expression changes in asthma exist. The intended user is a researcher who needs to audit a proposed classification benchmark before relying on its sample-random performance. The code provides original-source manifests, phenotype extraction, a study-disjoint split, a second within-study site split, test-set uncertainty and explicit refusal to merge incompatible probe ID spaces. It does not output patient diagnoses.

Study-level and site-level splits differ. BAMSE versus INMA isolates two cohorts processed on GPL17586, but the parent series and publication overlap. Wroclaw versus Zurich isolates locations within one independent series on GPL570, but not publication or laboratory processing. Neither split alone establishes generalization to a prospective hospital. The scopes remain separate in result files and throughout this report.

2. Dataset provenance and inclusion

The original gzip-compressed GEO matrices were downloaded directly from:

- GSE141623: https://ftp.ncbi.nlm.nih.gov/geo/series/GSE141nnn/GSE141623/matrix/GSE141623_series_matrix.txt.gz
- GSE141631: https://ftp.ncbi.nlm.nih.gov/geo/series/GSE141nnn/GSE141631/matrix/GSE141631_series_matrix.txt.gz
- GSE110551: https://ftp.ncbi.nlm.nih.gov/geo/series/GSE110nnn/GSE110551/matrix/GSE110551_series_matrix.txt.gz

data/manifest.json and data/external_site_results.json record source SHA-256 hashes. Preparation scripts read !Sample_geo_accession and exact !Sample_characteristics_ch1 fields, not inferred labels from signal. data/cohorts.json and data/site_cohort.json retain each original sample accession and selected probe values. Full source matrices total about 190 MB and are not committed; each can be re-downloaded from its URL and checked against the recorded digest. A matrix with hundreds of GSM columns is one original series download, not hundreds of separately fetched accession datasets. A separate accession-level check individually fetched 120 GSM metadata records (40 per series, balanced by class), verifying phenotype, platform and series against the matrices and saving URL/response hashes in data/gsm_metadata_audit.json. These metadata pages are not 120 expression datasets; no model changed.

BAMSE has 98 asthma-labeled samples and 158 controls, prevalence 0.3828. INMA has 36 cases and 165 controls, prevalence 0.1791. The third series has 78 asthma and 78 non-asthma adults. Wroclaw has 117 profiles (58 cases) and Zurich 39 (20 cases). These prevalence and recruitment differences are plausible sources of transfer failure. The GEO fields provide no independent clinical re-adjudication.

3. Fixed feature construction and leakage control

For each GPL17586 cohort, the parser takes the first 1,000 valid numeric probe rows in matrix order. Their exact intersection is 1,000 probe IDs. The choice is computational and independent of phenotype, not a biologically optimized gene signature. GPL570 likewise uses its first 1,000 valid rows, but there are zero probe IDs shared with the GPL17586 panel. The data object stores GEO sample IDs; overlap checks find zero repeated samples between BAMSE and INMA and zero between those cohorts and GSE110551.

Median imputation and standardization are inside a scikit-learn pipeline and fit only on the training study or site. Logistic regression has fixed $C=0.1$, $\text{max_iter}=1,500$ and seed 27. No features are selected by outcome on the test set. The platform difference is structural: an original model expecting the GPL17586 feature identity cannot be applied to GPL570 merely because both have 1,000 columns. Cross-platform gene mapping would require platform annotation, probe-to-gene disambiguation, common-gene aggregation and a prospective analysis plan; none was done here. The second experiment trains a new model on the independent platform rather than pretending the first was externally validated.

4. Outcomes and gate

The primary descriptive metric is held-out AUROC, the probability that a randomly chosen case scores above a randomly chosen control plus half of ties. Balanced accuracy at threshold 0.5 is secondary; it averages sensitivity and specificity. The build-time exploratory gate was $\text{AUC} > 0.65$ in both directions of a cohort pair. This threshold was scoped during the build and is not a third-party preregistration. Within-study random sample splits are diagnostic comparators only, never substituted for the holdout. Neural models were deliberately not tested after the baseline failed because fitting a more flexible model to these outcomes would invite outcome-driven tuning.

Train	Held out	n train	n test	Positive test	AUROC	Balanced accuracy	Conditional bootstrap 95% AUC interval
BAMSE GSE141623	INMA GSE141631	256	201	36	0.4571	0.4773	0.3505-0.5599
INMA GSE141631	BAMSE GSE141623	201	256	98	0.5213	0.5302	0.4477-0.5957
Wroclaw, GSE110551	Zurich, GSE110551	117	39	20	0.6132	0.5592	0.4237-0.7842
Zurich, GSE110551	Wroclaw, GSE110551	39	117	58	0.5447	0.5811	0.4392-0.6496

Both study directions and both site directions fail at point estimates. The random-sample AUCs within BAMSE and INMA were 0.5180 and 0.6055 respectively; these are diagnostics and not independent validation. No claim of a dramatic random-versus-study gap follows when one within-study AUC is itself weak. The most honest result is that this simple fixed panel fails, not that all asthma expression models must fail.

5. Uncertainty and sensitivity

`uncertainty.py` refits only the same fixed training pipelines to reproduce held-out scores. It then draws 2,000 class-stratified bootstrap samples of test subjects with replacement, without touching model parameters or the feature panel. Percentile intervals are listed above. This is conditional uncertainty for the chosen studies, phenotypes, frozen panel and model. It does not quantify platform harmonization error, selection of cohorts, label verification or a future site's case mix. The Zurich test interval is especially wide at $n=39$; its upper end crosses the exploratory threshold even though the point estimate and reverse direction fail. We do not reinterpret that upper interval as a passing gate.

The sample-random comparator is vulnerable to within-study structure even when no subject is duplicated. Conversely, study/site splits introduce severe distribution differences that may depress true signal or expose nontransportability. Distinguishing those explanations requires additional independent data and planned covariate analyses. No post-hoc stratification by age, sex or obesity was used to declare success.

6. Cohort search and exclusions

The independent-cohort search examined original GEO accession metadata. GSE141661 is the parent of BAMSE and INMA and cannot be counted as a third cohort. GSE137394 profiles asthma-at-baseline samples but lacks a non-asthma target group in the inspected series metadata. GSE188424 asks controlled versus uncontrolled asthma, which is a different endpoint. GSE110551 has explicit asthma/non-asthma and site labels, but its GPL570 probe IDs differ entirely from GPL17586. A further platform-targeted search checked [GSE69683](#), a separate whole-blood asthma/control cohort on GPL13158, and [GSE130499](#), an asthma/control series of bronchial epithelial and lavage cells on GPL4133. Both have a

plausible disease comparison, but neither supplies the GPL17586 blood probe-and-tissue match needed for a frozen-model test. [GSE134347](#) uses GPL17586 blood profiles but contrasts sepsis, non-infectious critical illness and health, not asthma versus non-asthma. These are metadata-level scope checks, not downloaded cohort matrices or evidence that no suitable cohort exists. These exclusion decisions prevent a superficially larger but invalid pooled validation. Different tissues, species, treatment-response endpoints or single-cell profiles would also need separate questions and model definitions.

7. Reproducibility

From the repository root, install NumPy, scikit-learn and pytest. Run `python3 projects/transcript_shift/prepare.py` to retrieve and parse the two GPL17586 matrices, compare the printed SHA-256 values to committed data/`manifest.json`, and run `python3 projects/transcript_shift/evaluate.py`. For the independent site audit, run `python3 projects/transcript_shift/external_site.py`. Run `python3 projects/transcript_shift/uncertainty.py` for conditional AUC intervals, `python3 projects/transcript_shift/study_report.py` for cross-check tables, and `python3 -m pytest -q projects/transcript_shift`. The scripts make no account-dependent or paid-service calls. The committed extracted JSON supports model reproduction without re-downloading source matrices, but original-data provenance should be checked when publishing.

8. Limitations and verdict

The two primary cohorts are study-disjoint but publication-linked. The third is publication-distinct but evaluated only within its own site split. Phenotypes come from GEO submitter metadata, not a harmonized clinical adjudication. First-in-matrix probes are arbitrary; the panel may be biologically uninformative. Age, geography, recruitment, expression processing, disease prevalence and site batches are not adjusted. AUROC does not measure calibration or patient utility. No gene mechanism, biomarker nomination or clinical diagnosis is warranted. The result is a reproducible negative benchmark and an auditor for easy leakage, with a clear path to a genuinely independent hospital/endpoint-matched validation if original compatible expression and phenotype data become available.

Source notes

The [GEO GSE141623](#), [GEO GSE141631](#) and [GEO GSE110551](#) series pages identify the parent datasets. Original matrix URLs and hashes above are the computation sources. [GSE141661](#), [GSE137394](#), and [GSE188424](#) are scope checks, not extra training or test data. Numeric results are reproducible from committed data/`results.json`, data/`external_site_results.json`, data/`uncertainty.json` and the scripts named above.

Independent Asthma BRIDGE assay, same-study random split only

A fourth original GEO series, [GSE285752](#), reports 954 whole-blood profiles in an array experiment distinct from the three earlier series. Its platform is GPL10558, not the GPL17586 probe space of the original BAMSE/INMA model; it cannot be called that model's external validation. We downloaded the original submitter metadata XLS and full ~685 MB compressed normalized expression matrix from the GEO supplementary URLs in data/`bridge_protocol.json`; the source SHA-256 digests and code are committed, while the large matrix is excluded from Git and can be fetched again. The metadata has 852 whole-blood asthma and 102 controls. Four sample rows lack IDAT filenames; there are no duplicated *nonmissing* filenames. The matrix has 1,535 total sample columns, each paired with a detection-p-value column. `bridge_side_study.py` joins the 954 whole-blood sample names exactly and takes the first 1,000 numeric ILMN probe rows without consulting labels.

A local descriptive protocol was written before computing the result: one seed-27 stratified 75/25 sample split, training-only imputation/standardization and fixed logistic C=0.1. Its 715 train and 239 test samples include 639 and 213 asthma cases respectively. Held-out AUROC is 0.9074; average precision 0.9872 against test prevalence 0.8912; balanced accuracy at threshold 0.5 is 0.6829. This is a **within-one-study random sample split** with extreme class imbalance, not an independent hospital holdout, not a matched published benchmark and not transfer of either fitted GPL17586 model. A post-result confounding audit of the original metadata found controls aged 23.95-31.50 years (median 25.90), versus cases aged 5.17-31.65 years (median 10.03). All 102 controls but only 102 of 852 cases lie within the shared 23.95-31.50 age range; sex also differs (controls 62 F/40 M, cases 351 F/501 M). On the *same frozen split*, an age-only logistic model scored AUC 0.9497, higher than the expression model, although age was not an input to that expression model. Post-hoc restriction of the original held-out predictions to the overlapping age range leaves only 42 subjects (16 cases) and gives expression AUC 0.6707; this tiny, selected subset is descriptive and not a fresh validation, age-adjusted performance claim or benchmark pass. The apparently high all-age random-split AUC may reflect developmental differences and recruitment composition, so it cannot be interpreted as disease-specific signal. It does not reverse the earlier negative cross-study AUCs or meet the two-direction >0.65 transfer gate. No clinical diagnostic claim or cross-platform model mapping follows. Exact probe/metadata source hashes, counts and refusal scope are in data/`bridge_result.json`.

Independent blood RNA-seq cohort eligibility, September 26

[GSE207751](#) is an original independent IMSA cohort of 56 whole-blood RNA-seq subjects, with 27,071 unique ENSG gene rows and exactly 56 metadata-matched expression columns with no missing/nonfinite cells, in original [metadata](#) and [normalized/batch-corrected gene-expression matrix](#). The metadata lists 19 moderate asthma, 23 severe asthma and 14 healthy samples, with age, sex and six batch values. Batches and labels are entangled: batch 2 has eight cases and no healthy control; batch 7 has three cases and six healthy controls. The gene-level RNA-seq features cannot be passed unchanged to the original GPL17586 microarray probe-level classifier. The released normalized matrix is already batch-adjusted, without a separately frozen training-only adjustment in this audit. No model transfer or within-cohort score was calculated; 56 patients are not 56 distinct downloaded biological cohorts. Metadata and matrix response hashes, counts and decision are in `data/gse207751_scope.json`. This improves phenotype/source eligibility review but leaves the original independent compatible-cohort gate unmet.